

Commerce XI Maths App

Exercise 3.3

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 12

Question 1 - Q1(i)

If w is a complex cube root of unity, show that $(2 - w)(2 - w^2)$ equals

Options

- A. 7
- B. 3
- C. 5
- D. 9

Correct answer: 7

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives 7.

Question 2 - Q1(ii)

If w is a complex cube root of unity, find $(2 + w + w^2)^3 - (1 - 3w + w^2)^3$

Options

- A. 65
- B. 64
- C. 63
- D. -65

Correct answer: 65

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives 65.

Question 3 - Q1(iii)

If w is a complex cube root of unity, find $(a + bw + cw^2)/(c + aw + bw^2)$

Options

- A. w^2
- B. w
- C. 1
- D. $-w^2$

Correct answer: w^2

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives w^2 .

Question 4 - Q2(i)

If w is a complex cube root of unity, find $w + (1)/(w)$

Options

- A. -1
- B. 1
- C. 0
- D. w^2

Correct answer: -1

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives -1.

Question 5 - Q2(ii)

If w is a complex cube root of unity, find $w^2 + w^3 + w^4$

Options

- A. 0
- B. 1
- C. -1
- D. 3

Correct answer: 0

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives 0.

Question 6 - Q2(iii)

If w is a complex cube root of unity, find $(1 + w^2)^3$

Options

- A. -1
- B. 1
- C. 0
- D. -8

Correct answer: -1

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives -1.

Question 7 - Q2(iv)

If w is a complex cube root of unity, find $(1 - w - w^2)^3 + (1 - w + w^2)^3$

Options

- A. 0
- B. 8
- C. -8
- D. 16

Correct answer: 0

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives 0.

Question 8 - Q2(v)

If w is a complex cube root of unity, find $(1 + w)(1 + w^2)(1 + w^4)(1 + w^8)$

Options

- A. 1
- B. 0
- C. -1
- D. 4

Correct answer: 1

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives 1.

Question 9 - Q3

If α and β are the complex cube roots of unity, find $\alpha^2 + \beta^2 + \alpha\beta$

Options

- A. 0
- B. 1
- C. -1
- D. 3

Correct answer: 0

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives 0.

Question 10 - Q4

If $x = a + b$, $y = \alpha a + \beta b$ and $z = a\beta + b\alpha$, then xyz equals

Options

- A. $a^3 + b^3$
- B. $a^3 - b^3$
- C. $(a + b)^3$
- D. $ab(a + b)$

Correct answer: $a^3 + b^3$

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives $a^3 + b^3$.

Question 11 - Q5(i)

If w is a complex cube root of unity, find $(w^2 + w - 1)^3$

Options

- A. -8
- B. 8
- C. -1
- D. 0

Correct answer: -8

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives -8 .

Question 12 - Q5(ii)

If w is a complex cube root of unity, find $(a + b) + (aw + bw^2) + (aw^2 + bw)$

Options

- A. 0
- B. $a + b$
- C. $3(a + b)$
- D. $-(a + b)$

Correct answer: 0

Explanation and steps

Use $w^3 = 1$, $1 + w + w^2 = 0$, $1/w = w^2$ and $1/w^2 = w$. Simplifying with these identities gives 0 .